

Priyadharshini Ramesh Kumar

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Professional Summary

Data scientist with hands-on experience across machine learning, computer vision, and applied research, spanning end-to-end pipeline design (data cleaning, model development, deployment) and cross-functional research collaboration. Skilled in Python-based ML/DL frameworks, geospatial and real-time data visualization, and LLM-based document processing pipelines, with a track record of communicating technical findings to non-technical stakeholders.

Skills

Languages: Python, SQL, R

ML / Deep Learning: PyTorch, Keras, HuggingFace Transformers, Scikit-learn, CNN, VGG16 (transfer learning), Optuna (hyperparameter tuning), GridSearchCV, Adam optimizer

Data Science & Statistics: t-SNE, KMeans, Bayesian Networks, Time-Series Diagnostics (ADF test, ARCH test), NumPy, SciPy

Computer Vision: OpenCV (cv2), scikit-image, Open3D, PyWavelets, Spectral, PCA, Depth Estimation (GLPN, DPT), Point Cloud Processing, Poisson Surface Reconstruction

NLP / LLMs: GPT-4o, Retrieval-Augmented Generation (RAG) Pipelines

Visualization & Web Tools: Streamlit, Leaflet, Plotly, Gradio, GeoJSON

Platforms: Google Colab (GPU), Hugging Face Spaces, Git/GitHub

Other: NetworkX, DBSCAN, Hierarchical Clustering, Arduino, IoT (GSM, RFID)

Experience

AI Intern - OPX AI (Startup)

Jan 2026 - Apr 2026

Houston, TX

- Built a multi-stage pipeline converting raw oil & gas technical PDFs into a structured, RAG-ready knowledge base using GPT-4o, processing 300+ document chunks per batch run.
- Designed a two-stage GPT-4o extraction process over ~20k-character document windows to detect operational scenarios and generate structured playbooks (trigger conditions, decision points, escalation paths) traceable to source file and page.
- Built a governance validator enforcing an 8-section compliance checklist via regex-based rules, auto-approving 256 of 483 chunks in a single batch, and implemented GPT-based domain taxonomy discovery with human-in-the-loop approval and full before/after change logging.
- Developed a three-layer PDF duplicate detector (byte hash, text fingerprint, GPT semantic comparison) and an atomic, collision-safe ID registry for corpus hygiene.

Research Data Scientist - Texas A&M University

Aug 2024 - Jan 2026

College Station, TX

- Implemented and benchmarked four lightning flash-stitching algorithms (sequential threshold-based, DBSCAN, hierarchical agglomerative, graph-based/NetworkX) to group raw GLM satellite detections into discrete flash events, following published spatiotemporal methodology.
- Quantitatively compared algorithm outputs via flash count, megaflash count, and extent distributions, measuring pairwise agreement using Adjusted Rand Index (e.g., 0.875 between DBSCAN and graph-based methods).
- Built a Streamlit/Leaflet tool parsing raw HLMA sensor exports (.dat/.dat.gz) with a fallback parser for malformed files, rendering altitude-tiered lightning clusters cross-referenced against NOAA storm-event reports.
- Performed data cleaning, preprocessing, and visualization support for the research group; prototyped FloodSense, an exploratory Streamlit flood-risk tool combining rainfall and infrastructure data.

Research Assistant - Texas A&M University, SGL Lab

Feb 2024 - Apr 2024

College Station, TX

- Conducted initial data discovery on global wood trade-flow data (import/export volumes by wood type, production drivers, downstream industries) prior to a pivot toward causal/probabilistic modeling.
- Built a 3-node Bayesian network analysis of a semiconductor supply chain (government funding to chip manufacturing to exports), discretizing variables via empirical CDF-based tercile thresholds and joint probability tables.
- Ran an Augmented Dickey-Fuller test (found the series non-stationary) and an ARCH test on regression residuals (found statistically significant heteroscedasticity, $p < 0.001$) to validate modeling assumptions.
- Communicated risk-propagation findings and modeling tradeoffs (non-stationarity, volatility clustering) to non-technical stakeholders.

Chennai, India

- Curated a labeled dataset of 26 Indian Sign Language gestures from simple- and complex-background video footage using Python and OpenCV, integrating extracted features into a larger, existing 41-class gesture recognition dataset.
- Built and trained a CNN and a VGG16 transfer-learning model for gesture classification, comparing architectures on accuracy, loss curves, and per-class precision/recall/F1.
- Part of a four-person student research team under faculty supervision, working toward an ISL recognition system for educational-domain signs, funded through the TIH-IoT CHANAKYA Fellowship (IIT Bombay's DST-sponsored IoT research hub).

Projects

Multimodal Fusion Network (Vision + Audio) - Python, Keras, Optuna, t-SNE

- Designed a dual-stream neural network pairing a three-layer Conv2D image encoder with a three-layer Conv1D audio encoder, fused through a dropout-regularized dense head into a 10-class classifier, achieving 98.9% accuracy and 98.9% F1 on validation.
- Jointly tuned both branches with Optuna across filter counts, dense units, dropout, and learning rate (best config: 48/128/64 image, 64/64/128 audio filters, 0.42 dropout); applied t-SNE and KMeans to fused embeddings to confirm class separation.

4D Thunderstorm Visualization - HLMA Capstone (Team of 2) - Python, Leaflet, Streamlit

- Built an interactive lightning visualization platform on Houston Lightning Mapping Array data (~150,000 new points per minute), replacing a static 2D map with altitude-based filtering, color-coded risk tiers, and zoom/pan interaction.
- Evaluated Bokeh, D3.js, and Leaflet, selecting Leaflet after prototyping and rejecting a 3D view for occlusion and rendering-cost issues; defined danger zones from lightning frequency and altitude thresholds. Supervised by Dr. Timothy Logan, Dept. of Atmospheric Sciences.

Air Pollutant Prediction from Multisensor Data (Team of 3) - Python, Scikit-learn

- Cleaned a 9,357-row UCI air-quality dataset to 6,095 usable rows, treating sensor error codes as missing, dropping a >90%-null column, and removing 846 outliers via a 1.4x IQR threshold.
- Benchmarked six regressors on RMSE and R^2 , tuning an MLP via 5-fold GridSearchCV to R^2 0.876; recommended Random Forest (R^2 0.872) over the marginally stronger MLP for interpretability and lower compute cost.

Hyperspectral Image Security Using LAB Color Space Encryption - B.E. Final Year Project, Anna University

- Designed a chaotic encryption algorithm (Pseudorandom Chaotic Image Scrambling) for hyperspectral images in CIELAB space, applying channel-independent pseudorandom permutation matrices across L, a, and b channels.
- Built a PCA and Gaussian-filtering preprocessing pipeline for the AVIRIS Indian Pines dataset (224 bands, 145x145 px), achieving 49.9 dB PSNR and 0.9998 SSIM on decrypted images.

DepthForge - Monocular 3D Reconstruction - Python, PyTorch, HuggingFace Transformers, Open3D, Gradio, Plotly

- Built an end-to-end pipeline converting a single 2D photo into a textured 3D mesh: depth estimation (GLPN or DPT), pinhole-camera point-cloud back-projection, statistical outlier removal, and Poisson surface reconstruction.
- Integrated two depth models with different tradeoffs (GLPN: indoor, 0.3-2.5s; DPT: indoor+outdoor, 3-8s) with a side-by-side comparison view; added automated mesh-quality metrics and privacy safeguards (consent gate, EXIF GPS detection); deployed on Hugging Face Spaces with PLY/OBJ/STL export.

Secure Smart Cabin Using Optimized Arduino GSM Interface (Team of 4) - Arduino, GSM, RFID, IoT

- Designed and built a smart-office IoT system on Arduino UNO integrating six modules: environmental sensing (DHT11, MQ-series), PIR motion detection, dual-mode secure entry (RFID + keypad), a monitoring dashboard, GSM/Wi-Fi connectivity (ESP32), and cloud storage (Adafruit IO, Arduino IoT Cloud).
- Implemented an IFTTT-based automation layer triggering multi-channel emergency alerts on smoke/fire detection; authored and published the system design as a peer-reviewed conference paper, later published as a Springer book chapter.

Education

Texas A&M University - M.S. in Data Science Aug 2023 - Dec 2024

GPA: 3.9/4.0

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Mining and Analysis, Statistical Computing in R & Python

Anna University, MIT Campus - B.E. Computer Science and Engineering Jul 2019 - Jul 2023

GPA: 8.78/10.0

Relevant Coursework: Big Data Analysis, Software Engineering, Operating Systems, Algorithms, Human Computer Interaction